

Homework 2 — Notebook and AI

Quantitative Reasoning II (INF 1220)

Due Wednesday, October 7, 11:59pm, on Populi

Work through the notebook and hold one study conversation with an LLM. Turn in both to Populi by the due date. Read *Using AI for Study and Data Analysis*, on the [Materials](#) page, before you start.

1. Setting up

You need a Google account. There is nothing to install.

1. Open the notebook: <https://colab.research.google.com/drive/1HB8LRiR1YsZzPRyYPgPQBnCfhLtAXpKF>
2. Click **File** → **Save a copy in Drive**. Do this before running or typing anything. The copy shared with you is read-only, so without this step nothing you do will save.
3. Its name will start with “Copy of”. Change this to include your last name.
4. Work in your copy. Any work you do in the original will not be saved.
5. Run the cells in order. **Shift** + **Enter** runs the cell you are in; **Runtime** → **Run all** runs everything.

Gemini is built into Colab for your use in coding. Hold the study conversation separately, at gemini.google.com. If you hold the study conversation within Colab, you won’t be able to share it.

2. Python Notebook

Answer the questions and do the tasks in the notebook.

The notebook uses Austin Airbnb listings from June 2026. It loads the data, describes the columns, and gives one worked example of reporting several regressions as one table. In this assignment, you will need to create the code to answer the questions (you can use Gemini to help).

Read *How to work in this notebook*, at the top of the notebook, before you start. In short:

- Every number should be produced by code. Tables, including regression tables, should be produced by code from your results, never typed by hand, and saved to a file (imagine you were going to add your tables to a paper).
- You can use Gemini to help you write the code, but you are responsible for the code you turn in.
- Use the `statsmodels` formula interface (`smf.ols`) for your models.

- The notebook must run top to bottom. Before you submit, choose **Runtime** → **Restart session and run all** and check that every cell runs without an error.

You are responsible for every number you report. Check the code before you trust its output.

3. LLM Study

Hold one conversation with an LLM covering these two topics:

1. Conditions for a valid confidence interval: random sampling, independence, sample size
2. Type II error and power: why “not significant” is not “no effect,” and how a study as large as *Heat and Learning* can detect a tiny effect

Follow the expectations in *Using AI for Study and Data Analysis*. Finish with the prompt from *Evaluate the conversation*, pasted as written, with those two topics as the list.

4. Assignment Submission

Turn in two items to Populi:

1. Your notebook, fully run. In Colab: **Share** → **Anyone with the link** → **Commenter**, then submit the link to the Populi assignment.
2. Your study conversation, shared the same way.

Points will be taken off if I try to open a link and I cannot see your work. Check both of your links in a private browsing window before you submit. Your notebook should be fully run, with all outputs saved.

Sharing an LLM conversation shares everything in the thread, so do not put anything private in the conversation you turn in.

5. If you do not use Colab or Gemini

You are not required to use Colab or Gemini (this is just the quickest set of tools to get spun up). If you choose to use another set of tools, (1) you are responsible for their functionality, and (2) upload your notebook to your Google Drive, change the sharing settings accordingly, and submit the link to the Populi assignment. Same goes for your LLM study conversation.